



INTRODUCTION

Multiple Stream Processes (MSPs) arise in manufacturing, healthcare, environmental monitoring, finance, cybersecurity, and other applications where several streams must be monitored simultaneously.

Despite substantial growth in MSP research, relatively few nonparametric monitoring methods have been developed for these settings, particularly for **detecting small and moderate process shifts**. Existing approaches often rely on distributional assumptions or asymptotic approximations that may be inappropriate during early monitoring stages.

The Cumulative Standardized Binomial EWMA (CSB-EWMA) control chart methodology addresses this gap by transforming observations relative to an in-control median and employing exact time-varying variance control limits for monitoring binary streams.

The csbewma package, now available on CRAN, implements the complete CSB-EWMA workflow, including optimal parameter selection, exact variance control charts, visualization tools, and post-hoc identification of potential out-of-control streams for MSP monitoring applications

METHODS

Table 1: Recoded Multiple Stream Process Data Structure

Sample Number				
Stream	j=1	j=2	...	j=t
1	x_{11}	x_{12}		x_{1t}
2	x_{21}	x_{22}		x_{2t}
$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
k	x_{k1}	x_{k2}		x_{kt}
Column Totals	C_1	C_2	...	C_t

For k independent streams, let $x_{it} \in \{0, 1\}$ be the binary indicator for stream i at time t .

The chart monitors:

$$W_t = \frac{Q_t - \mu t}{\sqrt{t\sigma^2}}, \text{ where } Q_t = \sum_{j=1}^t \sum_{i=1}^k x_{ij}$$

The EWMA recursion is:

$$r_t = \lambda W_t + (1 - \lambda)r_{t-1}$$

The exact variance at time t is:

$$Var(r_t) = \lambda^2 \left[2 \sum_{j=1}^{t-1} \sum_{i=j+1}^t (1 - \lambda)^{2t-i-j} * \frac{\sqrt{j}}{\sqrt{i}} + \sum_{i=1}^t (1 - \lambda)^{2t-2i} \right]$$

This formula enables time-varying control limits:

$$UCL_t = E[r_t] + L\sqrt{Var(r_t)}, LCL_t = E[r_t] - L\sqrt{Var(r_t)}$$

A signal is triggered when r_t falls outside UCL_t or LCL_t .

Note: Full derivations available in Reference 2.

Table 2: Optimal Parameters (λ, L)									
Target ARL ₀ = 370									
Range	0.1–0.2	0.2–0.3	0.3–0.4	0.4–0.5	0.5–0.6	0.6–0.7	0.7–0.8	0.8–0.9	0.9–1.0
Optimal λ	0.175	0.200	0.350	0.40	0.550	0.650	0.725	0.800	0.900
Optimal L	1.375	1.400	1.500	1.525	1.575	1.625	1.650	1.675	1.700
Achieved ARL ₀	372.32	375.21	373.90	374.48	368.83	373.83	372.99	367.53	360.11
Target ARL ₀ = 500									
Range	0.1–0.2	0.2–0.3	0.3–0.4	0.4–0.5	0.5–0.6	0.6–0.7	0.7–0.8	0.8–0.9	0.9–1.0
Optimal λ	0.150	0.200	0.300	0.450	0.525	0.650	0.700	0.850	0.925
Optimal L	1.525	1.575	1.650	1.725	1.750	1.800	1.800	1.875	1.875
Achieved ARL ₀	501.31	500.80	501.11	500.69	501.07	505.32	502.93	501.88	501.88

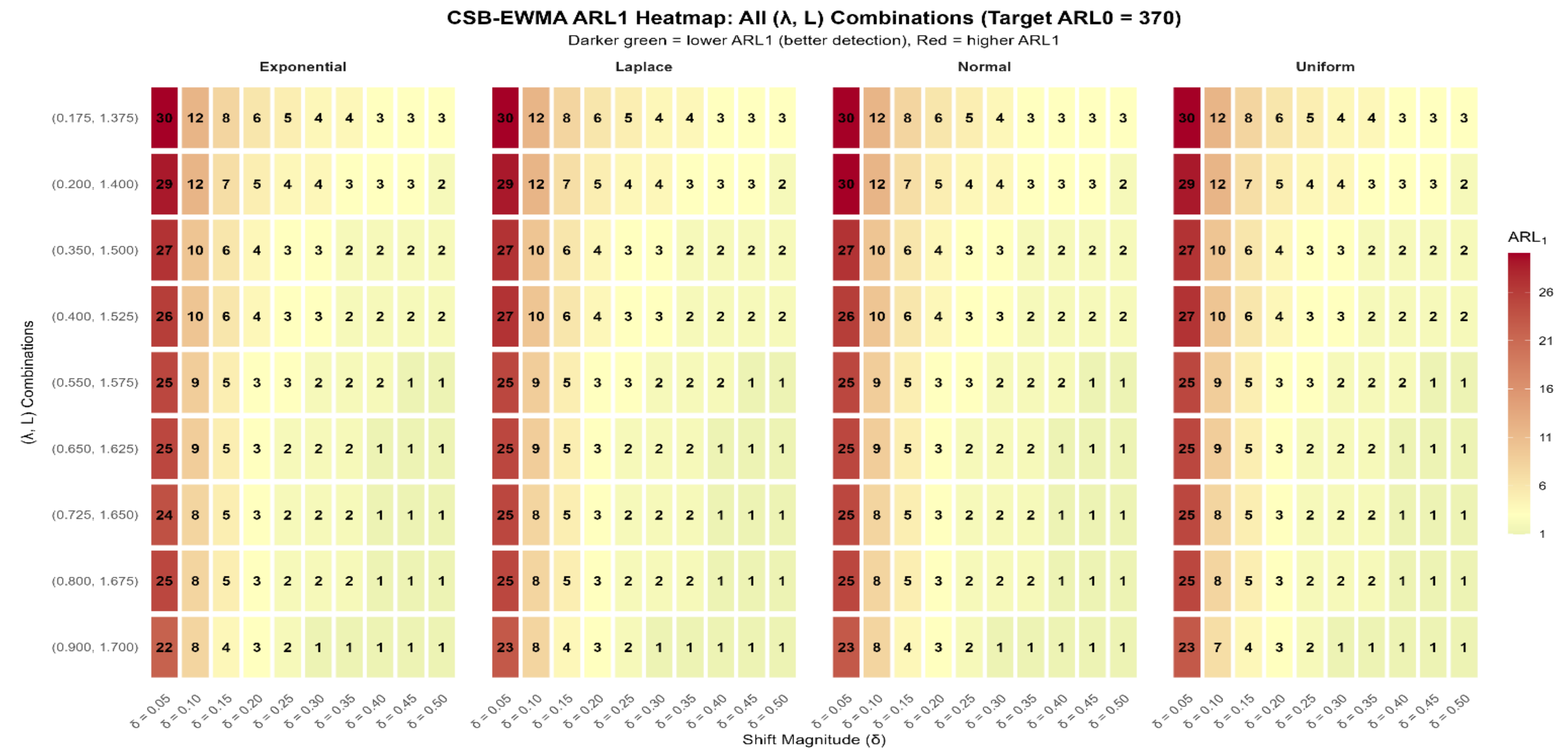


Fig. 1. Detection Capability of the Optimal Parameter

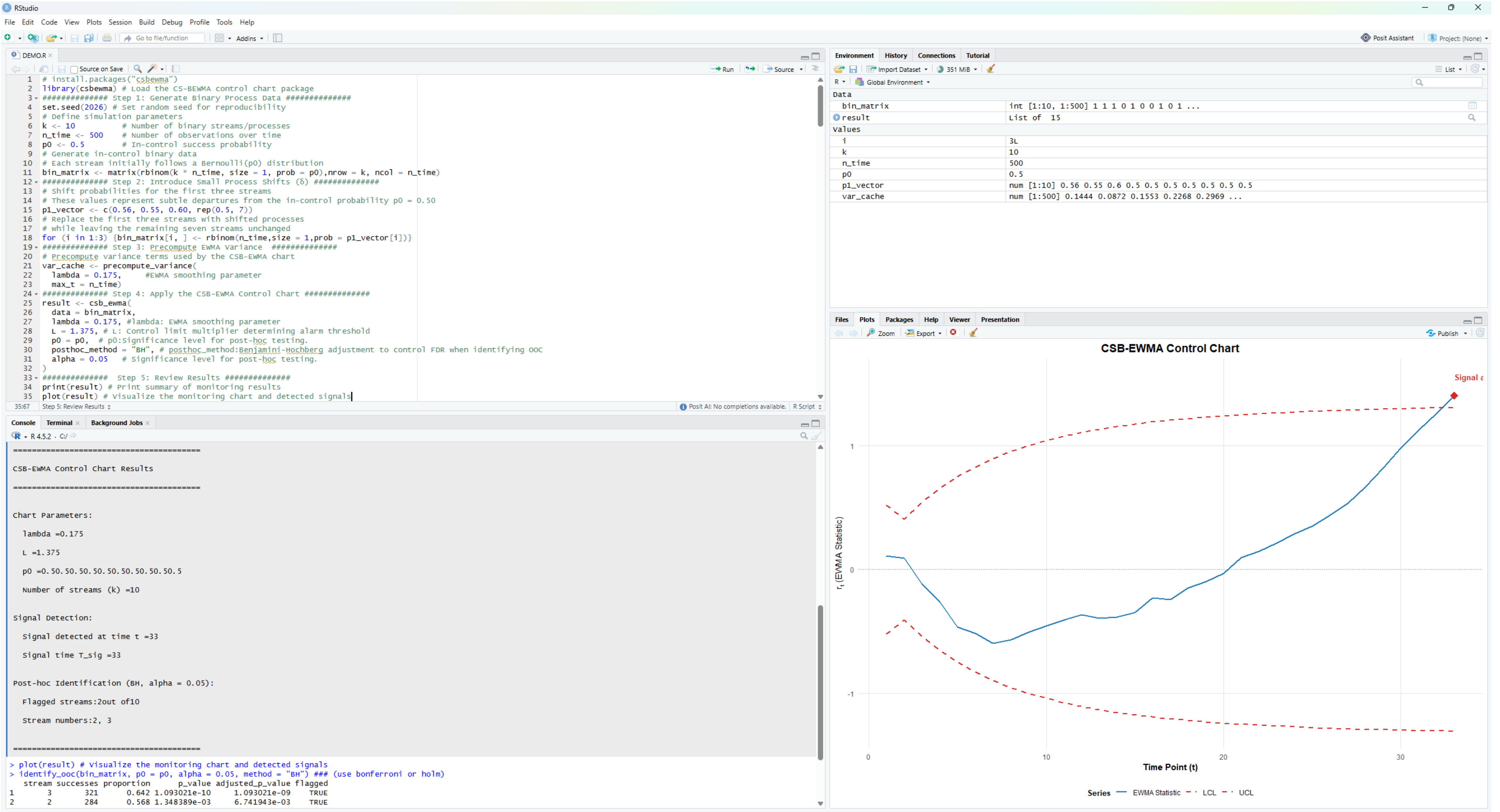


Fig. 2. Demo - Example csbewma Package workflow

DISCUSSION

Table 3: Post-Hoc Comparison (Full Dataset, 500 Time Points)

Stream	Phase II Proportion	Raw p-value	Bonferroni adj.	Holm adj.	BH adj
1	0.56	0.0088	0.0884	0.0708	0.0295
2	0.55	0.0013	0.0135	0.0121	0.0067
3	0.60	0.0000	0.0000	0.0000	0.0000
4	0.50	1.0000	1.0000	0.7292	0.6563
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
10	0.50	0.3437	1.0000	1.0000	0.5728

The simulation results demonstrate that the CSB-EWMA chart achieves the desired in-control performance while maintaining strong sensitivity to small and moderate shifts.

Table 2: shows that the optimized parameter combinations consistently achieve ARL₀ values near the target levels of 370 and 500 across a broad range of smoothing parameters.

Figure 1: highlights the detection performance is nearly identical across Exponential, Laplace, Normal, and Uniform distributions. This behavior is consistent with the nonparametric design of the CSB-EWMA methodology, which does not rely on distribution-specific assumptions.

- Across all tested distributions, ARL₁ decreases rapidly as shift magnitude increases. Small shifts ($\delta=0.05$) are detected within approximately 22–30 observations, while moderate-to-large shifts are often detected within one to three observations. This behavior demonstrates effective early-warning capability without sacrificing false-alarm control.

Figure 2: illustrates a typical workflow requiring only a few lines of R code to generate monitoring results and diagnostic plots using the CSB-EWMA package.

Post-hoc identification (identify_OOC()): The package includes a built-in function for identifying potential out-of-control streams after a signal occurs.

Table 3: demonstrates the corresponding output from identify_OOC() and illustrates the available multiple-testing adjustment options. Users may select *Bonferroni*, *Holm*, or *Benjamini–Hochberg (BH)* corrections depending on the desired balance between sensitivity and conservatism. Using the Figure 2 example, BH identifies all three shifted streams, whereas the more conservative Bonferroni and Holm adjustments identify Streams 2 and 3 only.

REFERENCES

- Muritala, F., Brown, A., Ghosh, D., & Ni, S. (2026). csbewma: Cumulative Standardized Binomial EWMA for Multiple Stream Processes (R package version 1.0.1). CRAN.
- Muritala, F., Brown, A., Ghosh, D., & Ni, S. (2026). Derivations for the Cumulative Standardized Binomial EWMA (CSB-EWMA) Control Chart. arXiv preprint arXiv:2601.09968.